

SHENQIU (HENRY) HUANG

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U.S. Permanent Resident | No sponsorship required | ITAR U.S. person

EDUCATION

Southern Methodist University — Lyle School of Engineering, Dallas, TX Expected May 2027

B.S. Electrical and Computer Engineering | Minor in Music (Saxophone) | GPA: 3.4/4.0

Embrey Academic Excellence Scholarship

Relevant Coursework: Circuit Design, Signal Processing, Microcontrollers & Embedded Systems, Digital Logic/VLSI Test, Semiconductor Fabrication

Marcus Air, Mesquite, TX March 2026 - Present

Student Pilot, Private Pilot License — 36 flight hours & solo endorsement

ENGINEERING EXPERIENCE

Beckhoff Automation GmbH & Co. KG Summer 2026

Engineering Intern

- Developed and troubleshot TwinCAT PLC/motion-control test systems, including laser-sensor/PID ball levitation, a G-code writing robot, and Windows-to-Linux HMI deployment.
- Built Python ADS handshake and buffer logic to transfer approximately 3,097 lines of G-code to a PLC; diagnosed state, communication, and repeatability issues through iterative testing.
- Integrated machine vision, 3x3 calibration, conveyors, multi-axis motion, and HMI into a sort/cut system; authored technical guides and validation procedures and built an offline installation package for end-user support.

SMU Lyle School of Engineering Jun 2025 – Sep 2025

Semiconductor Researcher & Laboratory Teaching Assistant

- Designed and tested digital logic in Siemens Tessent for defect detection and fault diagnosis; analyzed test methods and semiconductor screening reliability.
- Trained and guided 20 students through lab setup and equipment troubleshooting, explaining safe operating procedures and resolving hands-on technical questions.

SMU Physics Laboratory – CERN Experiment May 2024 – Aug 2024

Research Assistant, Instrumentation & Test Automation

- Automated optical-fiber stability and quality checks in Python/PyVISA across VISA, serial, and USB bench instruments, improving repeatability of lab measurement runs.
- Troubleshoot instrument communications and calibration workflows; maintained test inventory and documented procedures and results for continued laboratory use.

SELECTED ENGINEERING PROJECTS

77 GHz FMCW Radar Collision-Avoidance System Aug 2025 – May 2026

Texas Instrument Sponsored Senior Design

- Implemented real-time range, relative-velocity, and target-selection logic on TI's AWR2944P mmWave platform; validated system behavior across multiple road scenarios.

Aircraft Data & Communication 2023 – Present

Independent Projects

- Building an ESP32 ADS-B aircraft-traffic display; previously published a comparative study of UART, SPI, and I2C communication standards at the 2023 IEEE EEBDA Conference.

1995 Mazda Miata ECU Repair Summer 2025

Personal Electronics Project

- Diagnosed circuit faults, replaced and soldered components, and verified restored ECU operation.

TECHNICAL SKILLS

Schematic & Circuit Design: Schematic capture and circuit analysis (LTspice, AutoCAD, Fusion 360); board-level fault isolation and troubleshooting

Test Equipment & Instrumentation: Oscilloscope, bench-instrument automation and calibration, VISA/PyVISA-based test-equipment control, semiconductor test

Communication Protocols & Embedded: UART, SPI, I2C, serial, USB, ADS-B; TwinCAT 3, PLC, PID/motion control, Python, C/C++, ARM Assembly, ESP32

Documentation & Software: Technical/maintenance documentation, validation procedures, Beckhoff Xae, MATLAB, Verilog, Vivado, Siemens Tessent, OpenCV, MS Office